Environmental Data Science (2026), **xx**:xx 1–8

CAMBRIDGE UNIVERSITY PRESS

Application Paper

Evaluating Bias-Adjusted Climate Data for Extreme Heat Metrics in Agricultural Applications

Muhammad Taky Tahmid

School of Marine Science and Policy, University of Delaware, Newark, DE, 19716, USA. E-mail: tahmid@udel.edu.

Keywords: ERA5, Extreme Degree Days, Bias-Corrected Climate Data, CMIP6, Agricultural Impacts

Abstract

Bias adjustment is widely applied to climate model output to reduce systematic errors in simulated temperature and is commonly used in climate impact, loss, and damage assessments of crop yields. Existing evaluations of bias-adjusted data, however, primarily focus on mean climate conditions, despite the fact that many impact assessments rely on heat-exposure metrics that are driven by temperature extremes. This study evaluates how bias adjustment propagates into Extreme Degree Day (EDD) metrics, a class of nonlinear, impact-relevant temperature indicators widely used in climate and environmental economics to estimate crop yield responses and climate-related damages. Using ERA5 reanalysis as a benchmark, we compare EDDs computed from raw and bias-adjusted CMIP6 models (GFDL-ESM4, MPI-ESM1-2-HR, EC-Earth3) over Europe at the ADM1 level for the period 1994–2014. EDDs are calculated across seven temperature thresholds (0, 4, 8, 12, 28, 30, 32°C) relevant for agricultural impacts, aggregated using harvested-area-weighted averages for seven crops under rainfed and irrigated management, and summarized over crop-specific phenological periods. Results show that bias adjustment consistently improves agreement with reanalysis at low to moderate heat thresholds (EDD $\leq$ 12°C), with mean MAE reductions of 53–111 °C·days at EDD-0 and 22–56 °C·days at EDD-8 across the three models. In contrast, at extreme thresholds (EDD $\geq$ 28°C), bias-adjusted EDDs show near-zero or slightly negative skill (ΔSkill = -2 to -1 °C·days), indicating no improvement over raw model output. These findings highlight an important trade-off in bias-adjustment procedures and underscore the need for threshold-specific validation when applying bias-adjusted climate data in impact studies.

Impact Statement

Bias-adjusted climate data are routinely used to estimate future agricultural damages, yet their reliability for extreme heat metrics remains poorly understood. This study demonstrates that while bias adjustment improves moderate-threshold heat exposure metrics by 22–111 °C·days depending on model and threshold, it provides minimal or slightly negative improvement at the extreme thresholds (28–32°C) that drive nonlinear crop yield losses. Our findings have direct implications for climate impact assessments: researchers using bias-adjusted data for damage functions should validate performance at the specific thresholds their applications depend on, rather than assuming uniform improvement. By providing a multi-model, multi-crop, spatially explicit evaluation framework spanning seven crops and seven temperature thresholds across Europe, we offer guidance on when bias-adjusted data are trustworthy for agricultural heat stress applications and when caution is warranted.

1. Introduction

Bias-adjusted climate model outputs have become standard inputs for agricultural climate impact studies, providing temperature and precipitation data that are corrected to match observational climatologies

while retaining the climate model's representation of future change (Hempel et al 2013; Lange 2019). These datasets are widely used in crop modeling, climate-economy integrated assessment models, and climate loss-and-damage estimation frameworks (Lafferty and Sriver 2023; Mueller et al 2016). The motivation for bias adjustment is straightforward: raw climate model output often contains systematic biases in mean temperature, variability, and spatial patterns that can propagate into impact metrics, producing unrealistic estimates of crop yields, water demand, or economic damages (Maraun 2016).

However, validation of bias-adjusted data has typically focused on mean climate conditions—seasonal averages, monthly temperature distributions, or broad climatological patterns (Iturbide et al 2022). This practice is problematic for agricultural impact applications, which increasingly rely on derived metrics that respond nonlinearly to temperature extremes. Extreme Degree Days (EDD), for example, accumulate only the portion of temperature that exceeds crop-specific damage thresholds and are used extensively in empirical crop-climate response functions (Lobell et al 2013; Schlenker and Roberts 2009). Because EDD metrics are driven by the upper tail of the temperature distribution, small residual errors in mean state or variance after bias adjustment can translate into large errors in cumulative heat exposure (Di Luca et al 2020; Dosio 2016; Lafferty and Sriver 2023).

Crop yields respond nonlinearly to temperature, with sharp declines occurring once temperatures exceed critical thresholds during sensitive growth stages. For maize and other staple crops, yield losses accelerate rapidly above approximately 29–32°C during the reproductive period, leading to the widespread adoption of threshold-based heat metrics in the agricultural economics and agroclimate literature (Gourdji et al 2013; Lobell et al 2013; Schlenker and Roberts 2009). This creates a demanding test for climate data quality: if a bias-adjusted product underestimates the frequency or intensity of extreme heat days, EDD will be systematically too low, leading to underestimated climate damages.

Despite growing use of bias-adjusted climate data in agricultural damage assessment, systematic evaluations of how bias adjustment propagates into EDD metrics remain scarce. Existing validation studies focus on raw temperature distributions (Thorarinsdottir et al 2020) or on generic extreme indices (Dosio 2016; Iturbide et al 2022), but not on the crop-specific, threshold-aggregated, calendar-weighted heat exposure metrics that agricultural economists and impact modelers actually use in practice. This study addresses that gap by conducting a multi-model, multi-crop, spatially explicit evaluation of bias-adjusted CMIP6 temperature data for EDD calculation. Our central question is: when does bias adjustment improve EDD representation, and when does it degrade it?

2. Data and Methods

2.1. Study Region and Climate Data Sources

We analyze Europe at the first administrative subdivision level (ADM1), using administrative boundaries from the Global Administrative Areas database (GADM version 4.1). Our study period is 1994–2014, which is common to all three data sources and captures two decades of historical climate variability.

We compare three temperature datasets, all at daily temporal resolution: **ERA5 reanalysis** (Hersbach et al 2020) at 0.25° resolution, providing daily maximum ($T_{\max}$) and minimum temperature ($T_{\min}$) from hourly 2-meter air temperature fields; **Raw CMIP6 models** (GFDL-ESM4, MPI-ESM1-2-HR, EC-Earth3) from the historical experiment, obtained from the Pangeo CMIP6 archive at native resolutions of 0.7–1.0°; and **Bias-adjusted CMIP6** from the CIL-GDPCIR product via Microsoft's Planetary Computer, providing bias-corrected and downscaled data at 0.25° resolution using trend-preserving quantile mapping (Iturbide et al 2022; Lange 2019).

2.2. Extreme Degree Days Calculation

We calculate daily EDD using the sinusoidal approximation method (Snyder 1985), which models the diurnal temperature cycle as a sine wave. For a given day with $T_{\max}$, $T_{\min}$, and threshold T_{cutoff}: if

$T_{\max} \leq T_{\text{cutoff}}$, EDD = 0; if $T_{\min} \geq T_{\text{cutoff}}$, EDD = $(T_{\max} + T_{\min})/2 - T_{\text{cutoff}}$; otherwise:

$$\text{EDD} = \frac{(T_{\max} - T_{\min})\theta}{\pi} - \frac{(T_{\text{cutoff}} - T_{\min})}{\pi}\sqrt{1 - R^2} \tag{1}$$

where $R = (2T_{\text{cutoff}} - T_{\max} - T_{\min})/(T_{\max} - T_{\min})$ and $\theta = \arccos(R)$. We compute EDD at seven thresholds: 0, 4, 8, 12, 28, 30, and 32°C. Additional details are provided in Supplementary Methods S1.

2.3. Crop-Specific Aggregation

Gridded EDD values were aggregated to ADM1 units using harvested-area-weighted averages from GAEZ v4.0 (Fischer et al 2021). We analyzed seven crops (wheat, maize, rice, barley, potato, sugar beet, rapeseed) under both rainfed and irrigated management. We further subset EDD by crop-specific phenological periods from the SAGE Global Crop Calendar (Sacks et al 2010): *plant*, *between* (vegetative and reproductive growth), and *harvest*. Daily EDD was summed within each period, producing three seasonal values per crop-year-ADM1 combination. Full methodological details on calendar computation, harvested area weighting, and spatial aggregation are provided in Supplementary Methods S2–S5.

2.4. Evaluation Metrics

We define ΔSkill = MAE(raw) − MAE(biasadj). Positive values indicate bias adjustment reduces error; negative values indicate degradation. We also compute quantile errors ($Q_{90/95/99}$), tail mean error (mean EDD in top 10% of ERA5 distribution), and Kolmogorov-Smirnov statistics. All metrics are calculated at the ADM1-crop-period-threshold level, then averaged for presentation.

3. Results

3.1. Overall Bias-Adjustment Performance

Figure 1 shows ΔSkill across the seven EDD thresholds, three models, and seven crops. Bias adjustment consistently improves performance at moderate thresholds (EDD 0–12°C). At EDD-0, mean ΔSkill is 111 °C·days for GFDL-ESM4, 63 °C·days for MPI-ESM1-2-HR, and 53 °C·days for EC-Earth3. At EDD-8, improvements remain substantial: 56 °C·days (GFDL-ESM4), 30 °C·days (MPI-ESM1-2-HR), and 22 °C·days (EC-Earth3). Crop-specific patterns show that potato and maize exhibit the largest absolute improvements at moderate thresholds: for GFDL-ESM4, potato shows 148 °C·days improvement at EDD-0 and 76 °C·days at EDD-8, while maize shows 134 °C·days and 77 °C·days respectively (Supplementary Table S2).

In contrast, at extreme thresholds (EDD ≥ 28°C), bias adjustment shows minimal or slightly negative performance. At EDD-28, mean ΔSkill is −2 °C·days for GFDL-ESM4, −1 °C·days for both MPI-ESM1-2-HR and EC-Earth3. At EDD-30 and EDD-32, ΔSkill values hover near zero (−1 to 0 °C·days), indicating that bias-adjusted EDDs have similar or slightly larger errors than raw CMIP6 output at these extreme thresholds.

3.2. Spatial Patterns and Baseline Performance

Raw CMIP6 models exhibit MAE of approximately 154 °C·days at EDD-0 (averaged across models, regions, and crops), declining to near-zero at EDD ≥ 28°C as exceedance of these thresholds becomes rare across most of Europe. Spatial patterns (Figure 2) show that most ADM1 regions across northern and central Europe exhibit positive ΔSkill at EDD-8, while some southern European regions show smaller improvements or occasional slight degradation. This geographic variation may reflect baseline

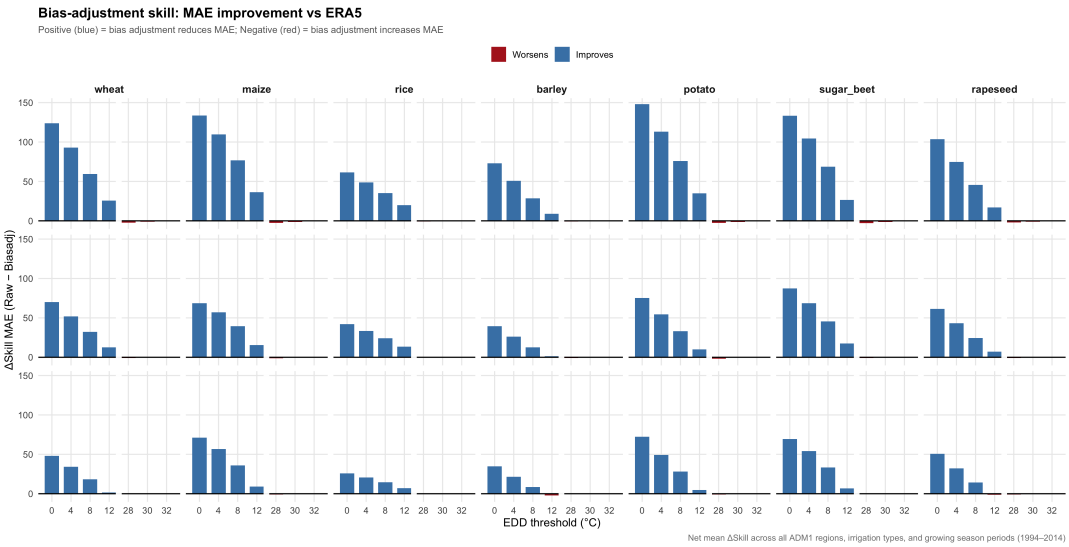

Figure 1. *ΔSkill (MAE improvement, in °C·days) for bias-adjusted vs. raw CMIP6 across EDD thresholds, crops, and models. Positive values (blue bars) indicate bias adjustment reduces MAE; negative values (red bars) indicate degradation. Bias adjustment consistently improves moderate thresholds (0–12°C) with MAE reductions of 22–111 °C·days (model-averaged), but provides minimal or negative improvement at extreme thresholds (≥28°C) where ΔSkill approaches zero..*

climate differences—regions with warmer climates may be more sensitive to tail-correction artifacts, while cooler regions benefit more from mean-state bias correction.

3.3. Quantile and Tail Behavior

Quantile error analysis (Figure 3) reveals that raw CMIP6 systematically underestimates EDD accumulation at Q90 and Q95 for moderate thresholds. For GFDL-ESM4 at EDD-0, raw Q90 error is −143 °C·days and Q95 error is −145 °C·days. Bias adjustment reduces these errors to 11 °C·days (Q90) and 18 °C·days (Q95), demonstrating effective correction of the cold bias in the upper-middle portion of the temperature distribution. However, at extreme thresholds (≥28°C), quantile errors for both raw and bias-adjusted data converge to small values, indicating minimal skill difference in the extreme tail.

Tail mean errors (Figure 4) show similar patterns. Raw CMIP6 exhibits large negative tail mean errors at EDD-0: −398 °C·days for GFDL-ESM4, −221 °C·days for MPI-ESM1-2-HR, and −201 °C·days for EC-Earth3. Bias adjustment reduces these errors to −154 °C·days (GFDL-ESM4), −140 °C·days (MPI-ESM1-2-HR), and −130 °C·days (EC-Earth3). However, at EDD-28 and beyond, both raw and bias-adjusted tail mean errors are small and similar in magnitude.

3.4. Distributional Fit

Kolmogorov-Smirnov statistics confirm that bias adjustment improves overall distributional fit at moderate thresholds but provides minimal benefit at extreme thresholds. Raw CMIP6 KS statistics at EDD-0 are 0.44 (GFDL-ESM4), 0.34 (MPI-ESM1-2-HR), and 0.33 (EC-Earth3). Bias adjustment reduces these to 0.18 for all three models, demonstrating improved distributional alignment. However, at EDD-28, raw KS statistics are 0.29 (GFDL-ESM4), 0.26 (MPI-ESM1-2-HR), and 0.23 (EC-Earth3), which bias

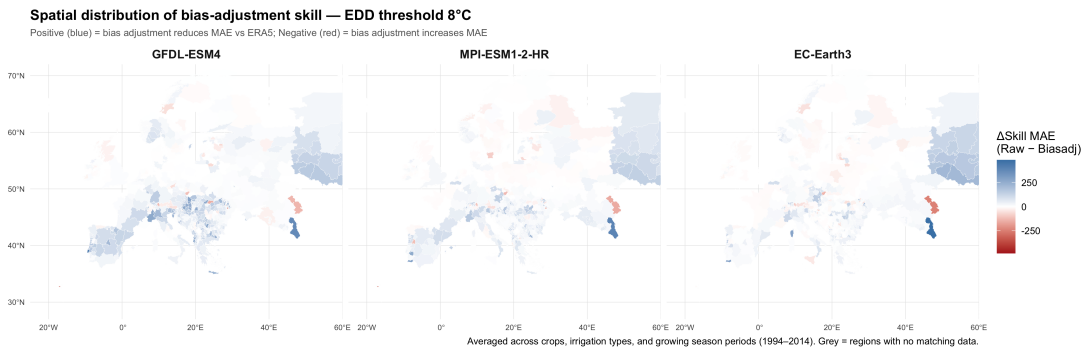

Figure 2. *Spatial distribution of ΔSkill (MAE improvement) at EDD threshold 8°C across European ADM1 regions for the three models. Blue indicates bias adjustment improves performance; red indicates degradation. Most of northern and central Europe shows substantial improvement, while southern Europe shows more mixed patterns. Gray regions indicate no matching data for that crop-model combination..*

adjustment reduces only marginally to 0.17–0.18, indicating minimal distributional improvement for extreme heat metrics.

4. Discussion

Our results reveal a systematic trade-off in bias-adjustment performance: bias-adjusted CMIP6 data reliably improve moderate-threshold EDD metrics (0–12°C), with mean MAE reductions of 22–111 °C·days, but show near-zero or slightly negative skill (−2 to −1 °C·days) at extreme thresholds (28–32°C). This dual pattern is consistent with known limitations of quantile-mapping correction methods, which can smooth or distort the tails of the distribution even as they correct the mean and central quantiles (Di Luca et al 2020; Lafferty and Sriver 2023; Lanzante et al 2018).

The improvement at moderate thresholds reflects successful correction of the raw CMIP6 cold bias in mean temperature. Because EDD at 0–12°C is driven primarily by the distribution body rather than the extreme tail, bias adjustment performs well in this regime. Agricultural applications that depend on baseline growing-season heat accumulation can rely on bias-adjusted data with confidence.

The near-zero skill at extreme thresholds is concerning from an impact-assessment perspective. EDD at 28–32°C is driven by rare, intense heat events in the warm tail—precisely the regime where bias-adjustment methods face their greatest challenges (Di Luca et al 2020; Dosio 2016; Iturbide et al 2022). Our findings align with those of Lafferty and Sriver (2023), who showed that bias-corrected climate data underestimate the largest weather-induced crop yield declines. The KS statistic results provide additional insight: at EDD-28, both raw and bias-adjusted data show similar distributional distance from ERA5 (KS ≈ 0.17–0.29), indicating that neither dataset reliably reproduces the full shape of the extreme heat distribution.

For agricultural impact studies, these findings have direct implications. Studies focused on moderate heat stress or temperature accumulation metrics below 15°C can use bias-adjusted CMIP6 data with confidence. However, impact studies that depend on damage thresholds above 28–30°C—common in agricultural economics and integrated assessment models (Moore and Diaz 2015; Schlenker and Roberts 2009)—should validate bias-adjusted data explicitly for those thresholds before use. Researchers face a

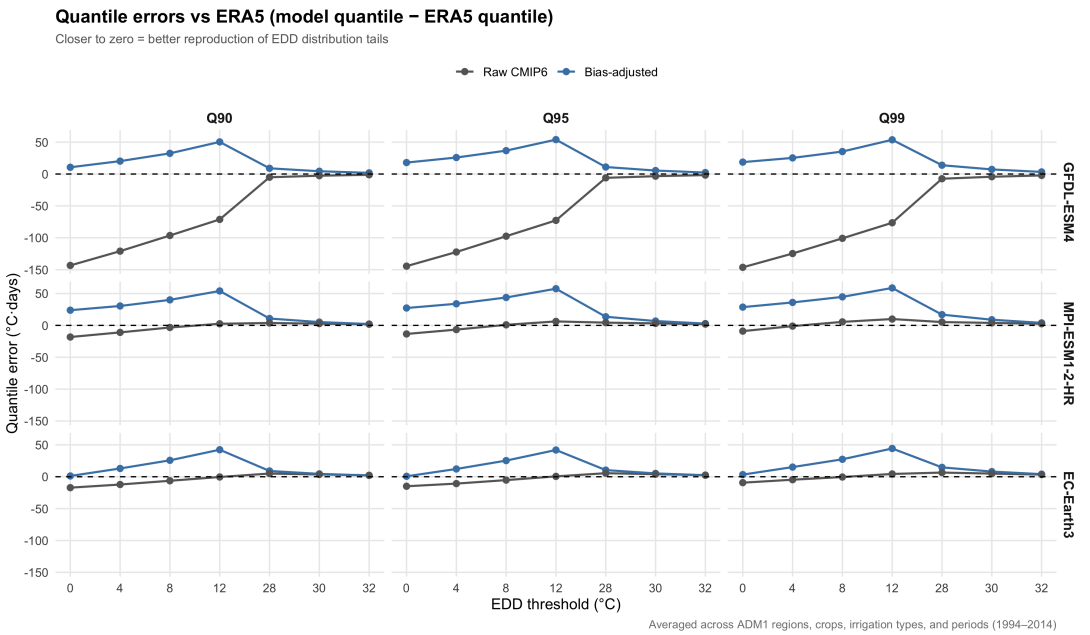

Figure 3. *Quantile errors (model quantile − ERA5 quantile, in °C·days) for raw and bias-adjusted CMIP6 at Q90, Q95, and Q99 across EDD thresholds. Raw CMIP6 (gray lines) shows large negative errors at Q90 and Q95 for moderate thresholds. Bias adjustment (blue lines) effectively corrects these errors at EDD 0–12°C. At higher thresholds (28–32°C), both raw and bias-adjusted errors converge to near-zero..*

choice: use raw CMIP6 output and accept mean-state biases, use bias-adjusted data with awareness of potential tail limitations, or use multi-model ensembles to capture structural uncertainty.

Our choice of ERA5 as the reference dataset introduces some uncertainty. While ERA5 is widely regarded as high-quality (Hersbach et al 2020), it is not equivalent to station observations for daily temperature extremes (Thorarinsdottir et al 2020). However, the CIL-GDPCIR product is explicitly corrected against ERA5, making this the most internally consistent reference choice.

We analyze only three CMIP6 models, though the consistency of threshold-dependent patterns across all three suggests that our findings are not model-specific artifacts but rather a general property of the bias-adjustment approach. Our analysis focuses on Europe; generalizability to tropical and subtropical climates—where extreme heat is more frequent—remains an open question. Future work should extend this analysis to additional regions, evaluate multiple bias-correction products, and assess whether bias-adjustment performance degrades further under projected warming when extreme heat becomes more common.

5. Conclusion

Bias-adjusted CMIP6 temperature data reliably improve moderate heat exposure metrics (EDD ≤ 12°C), with MAE reductions of 22–111 °C·days depending on model and threshold, but show near-zero or slightly negative skill improvement (−2 to −1 °C·days) at extreme thresholds (EDD ≥ 28°C) relevant for agricultural damage assessment. This threshold-dependent trade-off reflects a fundamental challenge in bias-correction methodology: methods designed to align mean climate with observations can fail to preserve the extreme tail behavior where rare but damaging heat events occur.

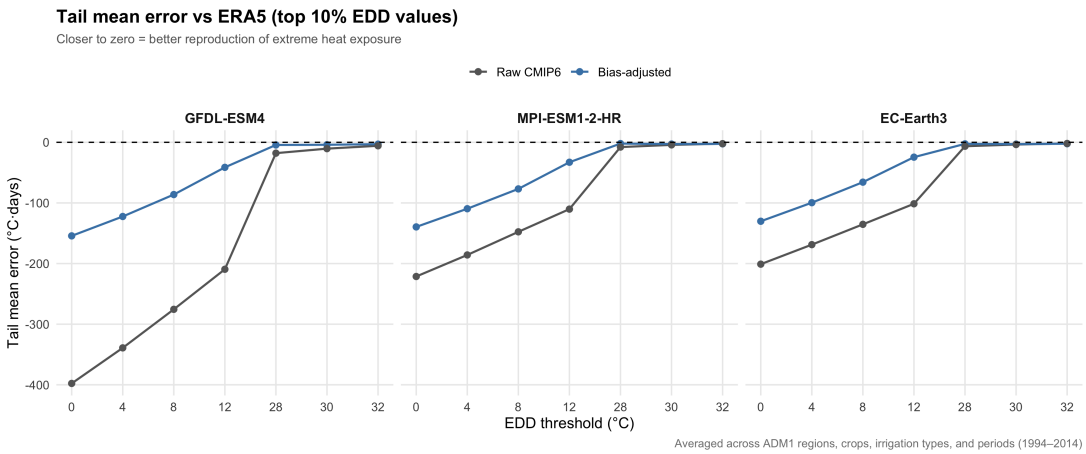

Figure 4. *Tail mean error (mean EDD in top 10% of ERA5 distribution, model – ERA5, in °C·days) for raw and bias-adjusted CMIP6 across EDD thresholds. Raw CMIP6 (gray lines) systematically underestimates extreme heat exposure at moderate thresholds. Bias adjustment (blue lines) reduces these errors substantially at EDD 0–12°C but provides minimal improvement at extreme thresholds (≥28°C)..*

For agricultural climate impact studies, these findings underscore the importance of metric-specific validation. Bias-adjusted data should not be treated as uniformly superior to raw model output; rather, their suitability depends on the thresholds and tail behavior that the impact application depends on. Our multi-model, multi-crop, spatially explicit evaluation across seven crops and seven temperature thresholds provides quantitative guidance on when bias adjustment helps and when it does not, supporting more informed data selection for agricultural impact assessment.

Acknowledgments. The author thanks Dr. James Rising for helpful discussions and feedback.

Competing Interests. The author declares no competing interests.

Data Availability Statement. Code and aggregated data are available at https://doi.org/10.5281/zenodo.19322211 and https://github.com/takyurp09/edd-bias-adjustment-europe. ERA5 data were obtained from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu). Raw CMIP6 data were accessed via Pangeo (https://pangeo.io). Bias-adjusted CMIP6 data (CIL-GDPCIR) were obtained from Microsoft Planetary Computer (https://planetarycomputer.microsoft.com). GAEZ v4.0 harvested area data are available from FAO (https://gaez.fao.org). SAGE global crop calendar data are available from Sacks et al (2010).

Author Contributions. M.T.T.: Conceptualization, Methodology, Data curation, Formal analysis, Visualization, Writing—original draft, Writing—review & editing.

References

Di Luca A, Pitman AJ and Elía R de (2020) Decomposing Temperature Extremes Errors in CMIP5 and CMIP6 Models. *Geophysical Research Letters* **47**, e2020GL088031. doi: 10.1029/2020GL088031.

Dosio A (2016) Projections of climate change indices of temperature and precipitation from an ensemble of bias-adjusted high-resolution EURO-CORDEX regional climate models. *Journal of Geophysical Research: Atmospheres* **121**, 5488–5511. doi: 10.1002/2015JD024411.

Fischer G, Nachtergaele F, Velthuizen H van, Chiozza F, Franceschini G, Henry M, Muchoney D and Tramberend S (2021) tech. rep. Global Agro-Ecological Zones v4 – Model documentation. Rome: FAO.

Gourdji SM, Sibley AM and Lobell DB (2013) Global crop exposure to critical high temperatures in the reproductive period: historical trends and future projections. *Environmental Research Letters* **8**, 024041. doi: 10.1088/1748-9326/8/2/024041.

Hempel S, Frieler K, Warszawski L, Schewe J and Piontek F (2013) A trend-preserving bias correction – the ISI-MIP approach. *Earth System Dynamics* **4**, 219–236. doi: 10.5194/esd-4-219-2013.

Hersbach H et al (2020) The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* **146**, 1999–2049. doi: 10.1002/qj.3803.

Iturbide M et al (2022) Implementation of FAIR principles in the IPCC: the WGI AR6 Atlas repository. *Scientific Data* **9**, 629. doi: 10.1038/s41597-022-01739-y.

Lafferty DC and Sriver RL (2023) Downscaling and bias-correction contribute considerable uncertainty to local climate projections in CMIP6. *npj Climate and Atmospheric Science* **6**, 158. doi: 10.1038/s41612-023-00486-0.

Lange S (2019) Trend-preserving bias adjustment and statistical downscaling with ISIMIP3BASD (v1.0). *Geoscientific Model Development* **12**, 3055–3070. doi: 10.5194/gmd-12-3055-2019.

Lanzante JR, Dixon KW, Nath MJ, Whitlock CE and Adams-Smith D (2018) Some pitfalls in statistical downscaling of future climate. *Bulletin of the American Meteorological Society* **99**, 791–803. doi: 10.1175/BAMS-D-17-0046.1.

Lobell DB, Hammer GL, McLean G, Messina C, Roberts MJ and Schlenker W (2013) The critical role of extreme heat for maize production in the United States. *Nature Climate Change* **3**, 497–501. doi: 10.1038/nclimate1832.

Maraun D (2016) Bias correcting climate change simulations - a critical review. *Current Climate Change Reports* **2**, 211–220. doi: 10.1007/s40641-016-0050-x.

Moore FC and Diaz DB (2015) Temperature impacts on economic growth warrant stringent mitigation policy. *Nature Climate Change* **5**, 127–131. doi: 10.1038/nclimate2481.

Mueller N et al (2016) Cooling of US Midwest summer temperature extremes from cropland intensification. *Nature Climate Change* **6**, 317–322. doi: 10.1038/nclimate2825.

Sacks WJ, Deryng D, Foley JA and Ramankutty N (2010) Crop planting dates: an analysis of global patterns. *Global Ecology and Biogeography* **19**, 607–620. doi: 10.1111/j.1466-8238.2010.00551.x.

Schlenker W and Roberts MJ (2009) Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change. *Proceedings of the National Academy of Sciences* **106**, 15594–15598. doi: 10.1073/pnas.0906865106.

Snyder R (1985) Hand calculating degree days. *Agricultural and Forest Meteorology* **35**, 353–358. doi: 10.1016/0168-1923(85)90095-4.

Thorarinsdottir T, Sillmann J, Haugen M, Gissibl N and Sandstad M (2020) Evaluation of CMIP5 and CMIP6 simulations of historical surface air temperature extremes using proper evaluation methods. *Environmental Research Letters* **15**, 124041. doi: 10.1088/1748-9326/abc778.